TOBE ENERGY CORP

FEED PACKAGE TRANSMITTAL

Zeeco ARC Hydrogen Electrolysis Facility

FEED PACKAGE TRANSMITTAL

TO: Rayford Forest, Zeeco Inc.
FROM: Colby DeWeese, Tobe Energy Corp
DATE: February 26, 2026
RE: FEED Package - Zeeco ARC Hydrogen Electrolysis Facility

Purpose

This package contains preliminary FEED-level documentation for the 12× T-25 hydrogen electrolysis installation at Zeeco's Advanced Research Complex (ARC) in Broken Arrow, Oklahoma.

The purpose of this transmittal is to support scope alignment discussions and enable Zeeco to proceed with facility planning, utility sizing, and integration engineering.

IFR Status: Issued for Review. Design subject to coordination, HAZOP, and utility verification.

Meeting Objective: Agree on battery limits and scope split; assign owners and due dates to all HIGH-priority open items.

Document Index

Primary meeting documents listed first.

Doc No.	Title	Rev	Status	Doc Owner	Section
TOBE-SCOPE-001	**Scope Split Matrix**	A	IFR	Both	4
TOBE-BL-001	**Battery Limits & Tie-Ins**	A	IFR	Both	4
TOBE-TBD-001	**Open Items Register**	A	IFR	Both	6
TOBE-PFD-001	Process Flow Diagram	A	IFR	Tobe	1
TOBE-PID-001	P&ID - Electrolyzer Skid	E	IFR	Tobe	1
TOBE-EL-001	Electrical One-Line Diagram	F	IFR	Tobe	2
TOBE-GA-001	General Arrangements	A	IFR	Tobe	1
TOBE-IO-001	I/O List - RO/DI Integration	A	IFR	Both	3
TOBE-STD-001	Codes, Standards & Design Basis	A	IFR	Tobe	5

Status Key: IFR = Issued for Review

Revision Log

Rev	Date	Description	By
A	2026-02-26	Initial issue for scope discussion	CFD

Distribution

Name	Company	Role
Rayford Forest	Zeeco	General Manager - ARC
Garrett Edler	Zeeco	Project Manager - ARC
Colby DeWeese	Tobe Energy	CEO / Technical Lead
Louis Mounsey	Tobe Energy	CFO / Commercial

Requested Actions

1. **Review** drawings and scope split for accuracy
2. **Confirm** utility tie-in points and capacities
3. **Identify** any gaps or conflicts with Zeeco facility plans
4. **Hold scope-split meeting** and freeze battery limits (target: March 2026)
5. **Confirm ownership** of: H_2 pipe to flare, O_2 vent stack, RO/DI interface wiring, E-stop stations, permitting lead

Contact

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EXECUTIVE SUMMARY

Zeeco ARC Hydrogen Electrolysis Facility

Project Overview

Parameter	Value
Location	Zeeco Advanced Research Complex, Broken Arrow, OK
System	12× Tobe T-25 Electrolyzers
Capacity	300 kg H_2/day (600 kW connected)
Phase 2	+1× T-125 (125 kg/day, 250 kW) - provisions only
H_2 Disposition	Direct to Zeeco flare/burners (no storage Phase 1)

Scope Summary

Zeeco Provides:

- 480V 3Φ utility service (65 kAIC assumed) + MCC-001 with space for Tobe feeder breaker
- Municipal water supply
- RO/DI water treatment system (Culligan) - including skid installation, drains, and reject routing
- Building/enclosure with HVAC (~10 kW cooling basis)
- H_2 separation, purification, and dehydration system (between electrolyzers and flare)
- H_2 piping from BL-H-01 through purification to flare - including supports, valves, and tie-in flange work
- Hydrogen flare system
- O_2 vent stack from BL-O-01 - including stack height/location per code and any dilution requirements
- Site civil, foundations, permits (AHJ lead)
- CAT6 network drop from facility switch to Tobe OT switch

Tobe Provides:

- 12× T-25 electrolyzer skids (factory-assembled)
- Feeder breaker CB-TOBE-001 (spec + furnish; Zeeco installs in MCC bucket)
- 480V feeder cable/conduit from MCC-001 to Tobe equipment
- 480V→5kV step-up transformer (XFR-001)
- 5kV switchgear and distribution
- Power electronics enclosures (EE-101, EE-102)
- Master PLC and HMI
- OPC-UA server + tag list + network requirements; Zeeco provides SCADA gateway/historian
- OT network switch and firewall (OT/IT boundary)
- Commissioning of Tobe equipment; Zeeco supports facility systems commissioning

Key Design Basis

Parameter	Value	Notes
Electrolyzer demand	576 kW	12 × 48 kW actual
480V auxiliary	~50 kW	RO, HVAC, controls
Total demand	~626 kW	
Transformer	750 kVA	85% loaded
5kV system	HRG, 600A, 25kAIC	High-resistance grounded
DI water consumption	~4,250 kg/day	10:1 water:H_2 ratio
Operating temp	<30°C	Near-ambient, minimal cooling
Enclosure HVAC	~10 kW total	2× units - by Zeeco
Water quality acceptance	<1 µS/cm	Per TOBE-IO-001

Battery Limits

ID	Service	Handoff Point	Tobe Provides	Zeeco Provides
BL-E-01	Electrical	MCC-001 load-side lugs	Breaker (spec/furnish), cable, conduit	MCC bucket space, install breaker
BL-W-01	DI Water	RO/DI skid outlet flange	Piping from flange to skids	RO/DI system, city water, isolation valve
BL-H-01	Hydrogen	2" 150# RF flange at skid	Skid piping to flange	Purification, dehydration, piping to flare
BL-O-01	Oxygen	3" 150# RF flange at skid	Skid piping to flange	Vent stack, height/routing per code
BL-C-01	Controls	Tobe OT switch port	OT switch, firewall, OPC-UA server	CAT6 drop to Tobe switch, SCADA gateway

Phase 1 vs Phase 2

Item	Phase 1	Phase 2
Electrolyzers	12× T-25	+1× T-125
Capacity	300 kg/day	+125 kg/day
Electrical	CB-5KV-03 space reserved in SWGR lineup	Install CB-5KV-03
Transformer	XFR-001 (750 kVA)	XFR-002 or service upgrade
Storage	None	TBD (compression, tanks)
Floor space	Space reserved for T-125 footprint (TBD)	Install T-125 + enclosure
Piping provisions	None (Phase 2 piping not included)	Add H_2/water tie-ins

Open Items (Summary)

1. **Utility service AFC** (Zeeco) - Confirm 480V available fault current (assumed 65 kAIC)
2. **Protection studies** (Tobe) - Coordination, arc flash, harmonics
3. **Safety architecture** (Both; driven by HAZOP) - PLC-based ESD vs dedicated safety PLC
4. **RO/DI integration** (Both) - Hardwired vs EtherNet/IP
5. **Building sizing** (Zeeco) - 3,000-4,000 sq ft recommended
6. **H_2/O_2 interface spec** (Both) - Pressure drop allowance, backpressure limit, material spec

Meeting outcome: Assign owners + due dates for all HIGH items in TOBE-TBD-001.

See **TOBE-TBD-001** for full register.

Schedule Alignment

Milestone	Target
Scope agreement	April 2026
Final FEED approval	May 2026
Equipment procurement	Q3 2026
Installation	Q4 2026
Commissioning	Q1 2027

I/O LIST - RO/DI SYSTEM INTEGRATION

TOBE-IO-001 | Rev A

Purpose

This document defines the instrumentation signals to be integrated from the Zeeco-provided Culligan RO/DI water treatment system into the Tobe master PLC.

Note: Tags reflect Culligan skid instrumentation naming; may be mapped to Tobe PLC tags during integration.

Analog Inputs (4-20mA)

Tag	Description	Range	Units	PLC Address	Notes
TT-101	Feed Water Temperature	32-120	°F	AI-001	Monitor inlet temp
FT-101	Feed Water Flow	0-50	GPM	AI-002	Totalize consumption
PT-101	RO Feed Pressure	0-150	psig	AI-003	Confirm if transmitter
PH-101	pH Analyzer	0-14	pH	AI-004	Quality verification
DO-101	Dissolved Oxygen	0-20	ppm	AI-005	Quality verification
EC-101	Conductivity	0-100	µS/cm	AI-006	Target <1 µS/cm
ORP-101	ORP Analyzer	-500 to +500	mV	AI-007	Redox potential

Total Analog Inputs: 7

Discrete Inputs

Tag	Description	Type	Normal	PLC Address	Notes
LS-101-LL	Feed Tank Low Level	N.C.	Closed	DI-001	Opens on low
LS-102-LL	DI Tank Low Level	N.C.	Closed	DI-002	Opens on low
P-101-RUN	Feed Pump Running	Aux	Open	DI-003	Closes when running
P-102-RUN	Transfer Pump Running	Aux	Open	DI-004	Closes when running
RO-101-FLT	RO System Fault	Relay	Open	DI-005	Closes on fault
RO-101-RDY	RO System Ready	Relay	Open	DI-006	Closes when ready

Total Discrete Inputs: 6

Discrete Outputs

Tag	Description	Type	Action	PLC Address	Notes
ESD-101	Inlet Shutoff Valve	Solenoid	Energize=Open	DO-001	De-energize to close
P-101-CMD	Feed Pump Start	Relay	Energize=Run	DO-002	
P-102-CMD	Transfer Pump Start	Relay	Energize=Run	DO-003	

Total Discrete Outputs: 3

I/O Summary

Type	Count
Analog Inputs (4-20mA)	7
Discrete Inputs (24VDC)	6
Discrete Outputs (24VDC)	3
Total I/O	**16**

Interface Options

Option 1: Hardwired (Preferred for Simplicity)

- Junction box at battery limit BL-W-01
- Multi-conductor cable to Tobe marshalling panel
- 4-20mA signals: shielded twisted pair
- Discrete: 24VDC common return

Option 2: EtherNet/IP (If Culligan PLC Available)

- Single CAT6 cable to Tobe network switch
- All signals via CIP messaging
- Reduces wiring, adds diagnostics
- Requires Culligan PLC configuration

Alarm & Interlock Summary

Condition	Setpoint	Action
LS-101-LL Low	Contact open	Alarm, inhibit P-101
LS-102-LL Low	Contact open	Alarm, inhibit electrolyzer start
EC-101 High	>5 µS/cm	Alarm, divert reject
PH-101 Out of Range	<6.0 or >8.5	Alarm
RO-101-FLT Active	Contact closed	Alarm
ESD Activated	Any ESD input	Close ESD-101, stop all pumps

BATTERY LIMITS & TIE-INS

TOBE-BL-001 | Rev A

Overview

Battery limits define the physical and functional boundaries between Tobe and Zeeco scope. All interconnections occur at these defined points.

Battery Limit Summary

ID	Service	Medium	Handoff Point	Tobe Side	Zeeco Side
BL-E-01	Electrical	480VAC 3Φ	MCC-001 load-side lugs	Breaker (spec/furnish), cable, conduit	MCC + bucket, utility, install breaker
BL-W-01	DI Water	Liquid	RO/DI outlet flange	Piping to skids	RO/DI system, isolation valve
BL-H-01	Hydrogen	Gas	2" 150# RF flange	Skid piping to flange	Sep/purif/dehy, piping, flare
BL-O-01	Oxygen	Gas	3" 150# RF flange	Skid piping to flange	Vent stack per code
BL-D-01	Drain	Liquid	Floor drain	Condensate lines	Drain system
BL-C-01	Controls	Data	Tobe OT switch port	OT switch, firewall, OPC-UA	CAT6 drop, SCADA gateway

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BL-E-01: Electrical Tie-In

Parameter	Value	Notes
Voltage	480VAC, 3Φ, 4W, 60Hz	Solidly grounded wye
Available fault current	65 kAIC	Verify with utility
Connected load	~650 kW	Electrolyzers + aux
Feeder breaker	CB-TOBE-001, 800AF	Tobe specifies + furnishes; Zeeco installs in MCC
Cable	TBD by NEC calc	By Tobe
Conduit	TBD	By Tobe
Handoff point	MCC-001 load-side lugs	Zeeco provides bucket space

Zeeco Action: Confirm MCC bucket space, AFC, and feeder provisions. Coordinate breaker installation with Tobe-furnished equipment.

BL-W-01: DI Water Tie-In

Parameter	Value	Notes
Medium	Deionized water	<1 μS/cm conductivity
Flow rate	~6 GPM nominal	~8 GPM peak
Pressure	20-60 psig	TBD by RO output
Temperature	Ambient	<90°F
Connection	1" flanged or union	TBD
Location	RO/DI skid outlet	

Zeeco Action: Confirm RO/DI outlet pressure, pipe size, connection type.

Signals at BL-W-01: See TOBE-IO-001 for instrumentation list.

BL-H-01: Hydrogen Tie-In

Parameter	Value	Notes
Medium	Hydrogen gas	>99.9% purity
Flow rate	~13 kg/hr nominal	300 kg/day ÷ 24
Pressure	<2 psig	Near atmospheric
Temperature	<40°C	Slightly above ambient
Connection	2" flanged	150# RF
Location	Electrolyzer skid outlet	

Key question: Does Zeeco flare accept near-atmospheric H_2 flow? Confirm backpressure limit at BL-H-01.

Zeeco Action: Confirm piping size to flare, material (SS required per B31.12), backpressure limit, and connection type.

BL-O-01: Oxygen Vent Tie-In

Parameter	Value	Notes
Medium	Oxygen gas	~99% purity
Flow rate	~100 kg/hr	8:1 mass ratio O_2:H_2
Pressure	Atmospheric	Vented
Temperature	<40°C	
Connection	3" flanged	150# RF
Location	Electrolyzer skid vent	

Zeeco Action: Confirm vent stack routing, height, and safety requirements.

BL-D-01: Drain Tie-In

Parameter	Value	Notes
Medium	Condensate / reject water	Non-hazardous
Flow rate	Intermittent	Low volume
Connection	Open drain to floor	
Location	Near electrolyzer skids	

Zeeco Action: Confirm floor drain locations and capacity.

BL-C-01: Controls Tie-In

Parameter	Value	Notes
Protocol	OPC-UA (primary)	EtherNet/IP also available
Physical	CAT6 Ethernet (shielded)	Zeeco provides drop to Tobe switch
Tobe provides	OT switch, firewall, OPC-UA server	+ tag list + network requirements
Zeeco provides	CAT6 cable, SCADA gateway, historian	
Handoff point	Tobe OT switch port	

Zeeco Action: Confirm SCADA integration requirements, IP addressing, and firewall rules.

Interface Responsibility Matrix

Tie-In	Pipe/Cable	Isolation Valve	Flange/Conn	Termination	Handoff
BL-E-01	Tobe (cable/conduit)	N/A	N/A	Tobe	MCC load-side lugs
BL-W-01	Tobe (from flange)	Zeeco	TBD	Both	RO/DI outlet flange
BL-H-01	Tobe to flange / Zeeco beyond	Tobe	Tobe	Both (witness)	2" 150# RF flange
BL-O-01	Tobe to flange / Zeeco beyond	Tobe	Tobe	Both (witness)	3" 150# RF flange
BL-D-01	Tobe	N/A	N/A	Tobe	Floor drain
BL-C-01	Zeeco (CAT6 drop)	N/A	N/A	Both	Tobe OT switch port

SCOPE SPLIT MATRIX

TOBE-SCOPE-001 | Rev A

Electrical

Item	Tobe	Zeeco	Notes
480V utility service		✓	65 kAIC assumed
Main breaker CB-001		✓	800AF, LSIG
Metering MTR-001		✓	kW/kVAR/PF/THD
MCC-001		✓	480V, 65kAIC
Feeder CB-TOBE-001	✓		Tobe spec/furnish; Zeeco installs in MCC bucket
480V→5kV transformer XFR-001	✓		750 kVA
5kV switchgear SWGR-5KV-001	✓		HRG, 600A, 25kAIC
5kV distribution	✓		To PE enclosures
Power electronics enclosures	✓		EE-101, EE-102
Electrolyzer skids	✓		12× T-25
Cable/conduit (480V feeder)	✓		From MCC to Tobe
Cable/conduit (5kV)	✓		Internal to Tobe scope
Grounding grid		✓	Per IEEE 80
Equipment grounding	✓		Tobe equipment only

Mechanical / Process

Item	Tobe	Zeeco	Notes
City water supply		✓	To RO/DI skid
RO/DI water treatment		✓	Culligan skid
DI water piping to electrolyzers	✓		From BL-W-01
Electrolyzer skids	✓		Factory-assembled
H_2 piping (skid to flange)	✓		To BL-H-01
H_2 separation / purification / dehydration		✓	Between electrolyzers and flare
H_2 piping (flange to flare)		✓	From BL-H-01 through purification
Hydrogen flare system		✓	Combustion, controls, stack
O_2 vent (skid to flange)	✓		To BL-O-01
O_2 vent stack		✓	From BL-O-01
Condensate drain	✓		To floor drain
Floor drains		✓	
Gas detection (enclosure)	✓		H_2 sensors in EE-xxx
Gas detection (building)		✓	If required by code

Controls & Instrumentation

Item	Tobe	Zeeco	Notes
Master PLC (PLC-101)	✓		System integration
HMI	✓		Local and/or remote
UPS for controls	✓		10 kVA
Network switches (Tobe)	✓		OT network
Network switches (facility)		✓	IT/SCADA network
Firewall (OT/IT boundary)	✓		Tobe default; confirm with Zeeco IT (see TBD #19)
SCADA gateway		✓	OPC-UA from Tobe PLC
RO/DI instrumentation		✓	Per TOBE-IO-001
RO/DI signal integration	✓		Pull into Tobe PLC
ESD logic	✓		Tobe PLC or safety PLC
E-stop stations	SHARED	SHARED	Zeeco: stations + wiring; Tobe: logic mapping + PLC inputs

Civil / Structural

Item	Tobe	Zeeco	Notes
Building/enclosure		✓	3,000-4,000 sq ft recommended
HVAC		✓	10 kW cooling (2× units)
Foundations		✓	Per Tobe GA drawings
Equipment pads		✓	
Fire suppression		✓	If required
Lighting		✓	
Access doors		✓	Personnel + equipment
Crane/rigging provisions		✓	For equipment install

Services

Item	Tobe	Zeeco	Notes
Engineering (Tobe equipment)	✓		
Engineering (facility integration)		✓	
Permitting		✓	Building, electrical, H_2
Equipment delivery to site	✓		
Rigging/setting equipment		✓	Or Tobe, TBD
Electrical terminations (Tobe)	✓		
Electrical terminations (Zeeco)		✓	
Piping connections	SHARED	SHARED	Zeeco: pipefitters; Tobe: flange kits + torque specs; both witness leak test
Commissioning (Tobe systems)	✓		
Commissioning (facility)		✓	
Training	✓		Tobe equipment operation
Warranty (Tobe equipment)	✓		Per contract

Legend

Symbol	Meaning
✓	Responsible party
SHARED	Joint responsibility, details TBD
(blank)	Not responsible

CODES, STANDARDS & DESIGN BASIS

TOBE-STD-001 | Rev A

Project Design Basis

System Configuration

Parameter	Value	Notes
Electrolyzer Model	T-25 (×12)	600 kW total
H_2 Production	300 kg/day	At 100% utilization
Operating Pressure	<2 psig	Near atmospheric
H_2 Purity	≥99.9995%	Fuel cell grade
Phase 1 Disposition	Flare	No storage

Operating Conditions

Parameter	Design Value	Reference
Operating Temperature	<50°C (122°F)	Near-ambient, isothermal design
Ambient Design Range	-15°C to +45°C	Standard; extended available
DI Water Quality	<0.1 µS/cm	After integrated RO/DI
Water Consumption	9-10 L DI/kg H_2	Net to stack
Specific Energy	~42 kWh/kg H_2	System, AC input
Electrical Efficiency	~94% HHV / ~79% LHV	Per Tobe PDG v1.1

Electrical Characteristics

Parameter	Value	Notes
Nominal Voltage	480V 3Φ 60Hz	±10% tolerance
Power Factor	0.95-0.99 leading	Capacitive load
THD (Current)	<5%	IEEE 519 compliant
THD (Voltage)	<3%	At PCC
Inrush Current	<1.2× FLA	LLC soft-start
Cold Start Time	<60 seconds	No thermal warmup

Materials

Component	Material	Notes
Electrolyzer Stack	304 Stainless Steel	No PGMs, membrane-free
H_2 Process Piping	316L SS	Per ASME B31.12
Structural	Carbon Steel	Galvanized where exposed

Primary Codes (AHJ Adopted)

Code	Title	Edition	Application
NFPA 2	Hydrogen Technologies Code	2023	Primary code for H_2 systems
NFPA 55	Compressed Gases and Cryogenic Fluids	2023	Storage quantities, separation
NFPA 70	National Electrical Code (NEC)	2023	Electrical installation
IBC	International Building Code	2021	Building/structural
IFC	International Fire Code	2021	Fire protection, hazmat
IMC	International Mechanical Code	2021	Mechanical systems

Note: Confirm AHJ-adopted editions during pre-application meeting with City of Broken Arrow.

Equipment & Design Standards

Electrolyzer & Hydrogen Generation

Standard	Title	Applicability
ISO 22734	H_2 generators using water electrolysis	Primary product standard - certification in progress Q2-Q3 2026
EIGA DOC 246	Guideline for Small Scale H_2 Production	Facility design reference

Piping & Pressure Equipment

Standard	Title	Applicability
ASME B31.12	Hydrogen Piping and Pipelines	**Primary H_2 piping code** - all H_2-containing piping
ASME B31.3	Process Piping	Water, N_2, instrument air piping
ASME BPVC VIII	Pressure Vessels	All pressure vessels >15 psig
ASME B16.5	Pipe Flanges and Flanged Fittings	Flange ratings

Hydrogen Quality & Purity

Standard	Title	Applicability
SAE J2719	Hydrogen Fuel Quality for Fuel Cell Vehicles	Purity specification reference
ISO 14687	Hydrogen Fuel Quality	International purity standard
CGA G-5.3	Commodity Specification for Hydrogen	Purity grades

Safety & Controls Standards

Process Safety

Standard	Title	Applicability
IEC 61511	Functional Safety - Safety Instrumented Systems	SIS design (SIL 2 target)
ISA 84	Safety Instrumented Systems	US adoption of IEC 61511
ISO/TR 15916	Basic Considerations for Safety of H_2 Systems	Safety philosophy reference

Hazardous Area Classification

Standard	Title	Applicability
IEC 60079-10-1	Classification of Areas - Explosive Gas	Zone classification
NFPA 497	Class I Hazardous Locations	Division classification
API RP 500	Classification of Locations for Electrical	Petroleum industry reference

Gas Detection

Standard	Title	Applicability
ISO 26142	Hydrogen Detection Apparatus - Stationary	H_2 detector specification

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Electrical Standards

Standard	Title	Applicability
IEEE 519	Harmonic Control in Electric Power Systems	Power quality, THD limits
IEEE 1547	Interconnection of DER with EPS	Grid interconnection (if applicable)
UL 508A	Industrial Control Panels	Panel listing
NFPA 79	Electrical Standard for Industrial Machinery	Machine electrical

Structural & Civil

Standard	Title	Applicability
ASCE 7-22	Minimum Design Loads and Associated Criteria	Load combinations
IBC 2021	International Building Code	Building design
ACI 318	Building Code for Structural Concrete	Foundations

Risk Category: III (per IBC Table 1604.5 - facilities containing hazardous materials)

Cybersecurity

Standard	Title	Applicability
ISA/IEC 62443	Industrial Automation and Control Security	Control system security

Anticipated / Emerging Standards

Standard	Status	Expected	Relevance
ISO 22734:2024	Published	Current	Updated electrolyzer standard - verify alignment
UL 2267	In development	2026	Equipment for hydrogen fuel production
NFPA 2 (2026)	Revision	2026	Check for amendments affecting separation distances
45V Final Rule	Pending	2025	Tax credit eligibility - impacts CI documentation
ASME B31.12 Addenda	Periodic	Ongoing	H_2 piping updates

Tobe Design Advantages (Compliance Impact)

Feature	Benefit	Standards Impact
Near-ambient operation (<50°C)	No high-temp insulation, simplified materials	Reduces B31.12 material requirements
Membrane-free design	No MEA degradation, no PGM procurement	Simplified supply chain
Caustic-free electrolyte	No chemical containment, no HAZMAT placarding	Eliminates secondary containment per IFC
Capacitive load behavior	Low THD, grid-friendly	IEEE 519 compliance inherent
SIL 2-capable SIS	Integrated safety system	IEC 61511 compliance

OPEN ITEMS REGISTER

TOBE-TBD-001 | Rev A

Summary

Priority	Count
HIGH	9
MEDIUM	7
LOW	4
Total	**20**

HIGH Priority

#	Item	Owner	Due	Status	Resolution / Next Step
1	**Confirm 480V available fault current** - Design assumes 65 kAIC. Verify with utility study.	Zeeco	TBD	OPEN	
2	**Protection coordination study** - Required before final breaker/fuse sizing.	Tobe	TBD	OPEN	
3	**Arc flash study** - Required for PPE labeling and ZSI settings.	Tobe	TBD	OPEN	
4	**Safety architecture** - PLC-based ESD vs dedicated safety PLC (SIL rated). Confirm with HAZOP.	Tobe	TBD	OPEN	
5	**RO/DI signal availability** - Confirm Culligan provides all signals per TOBE-IO-001.	Zeeco	TBD	OPEN	
6	**Building/enclosure sizing** - Recommend 3,000-4,000 sq ft. Confirm layout fits Zeeco plans.	Zeeco	TBD	OPEN	
7	**H_2 piping material/routing** - Confirm SS pipe from BL-H-01 to flare.	Zeeco	TBD	OPEN	
8	**Permits** - Confirm H_2 system permit requirements (AHJ).	Zeeco	TBD	OPEN	
9	**H_2/O_2 vent interface spec** - Define pressure drop allowance, backpressure limit at BL-H-01, O_2 vent stack height/dilution, material spec (SS), tie-in location.	Both	TBD	OPEN	

MEDIUM Priority

#	Item	Owner	Due	Status	Resolution / Next Step
10	**Harmonics study (IEEE 519)** - Required for utility compliance.	Tobe	TBD	OPEN	
11	**DC protection scheme** - Fast fuses vs DC breaker for 7kVDC bus.	Tobe	TBD	OPEN	
12	**UPS runtime** - 30 min assumed. Confirm critical loads and duration.	Tobe	TBD	OPEN	
13	**SCADA integration** - Confirm OPC-UA tags, polling rate, historian.	Zeeco	TBD	OPEN	
14	**E-stop locations** - Joint layout required.	Both	TBD	OPEN	
15	**Equipment rigging** - Confirm crane/forklift access for installation.	Zeeco	TBD	OPEN	
16	**Firewall ownership** - Confirm OT/IT boundary firewall provided by Tobe or Zeeco IT.	Both	TBD	OPEN	

LOW Priority

#	Item	Owner	Due	Status	Resolution / Next Step
17	**EMC/shielding plan** - Power/control cable segregation details.	Tobe	TBD	OPEN	
18	**5kV insulation clearances** - Confirm per IEEE/IEC.	Tobe	TBD	OPEN	
19	**Transformer inrush** - Precharge scheme if needed.	Tobe	TBD	OPEN	
20	**Spare parts list** - Develop for commissioning.	Tobe	TBD	OPEN	

Phase 2 Items (Future)

#	Item	Notes
P2-1	T-125 electrolyzer addition	+250 kW; CB-5KV-03 + floor space reserved in Phase 1
P2-2	XFR-002 or service upgrade	If load exceeds 750 kVA
P2-3	H_2 storage and compression	TBD scope and location

Resolution Process

1. Items assigned to **Owner** for resolution
2. Owner provides response/information
3. Both parties agree on resolution
4. Item marked **CLOSED** with resolution note
5. Register updated and re-issued

Change Log

Rev	Date	Changes
A	2026-02-26	Initial issue

TOBE-TBD-001 | Rev A | 2026-02-26